

Intrinsic Defect Self-Compensation Enables Spectrum Narrowing of Uniform Alloyed Ag–In–Ga–S Quantum Dots

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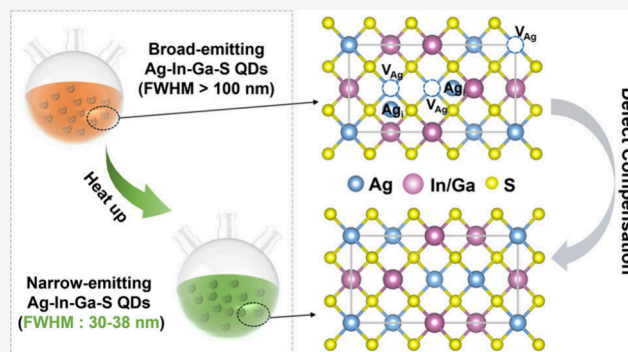


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Supporting Information

ABSTRACT: Eco-friendly Ag–In–Ga–S (AIGS) quantum dots (QDs) are promising alternatives to toxic Pb/Cd-containing QDs due to their tunable emission, high photoluminescence quantum yield (PLQY), and solution processability. However, their intrinsic defect-driven broad emission limits applications in high-resolution displays. Herein, we report a synthesis of uniform alloyed AIGS QDs with narrow emission via defect self-compensation, circumventing intricate shell growth. Benefiting from the defect compensation of interstitial Ag (Ag_i) to Ag vacancy (V_{Ag}) under thermal activation, structural defects are significantly suppressed, achieving tunable emission with remarkably narrow full width at half-maximum (fwhm = 30–38 nm) and high PLQYs. The resulting quantum dot light-emitting diodes (QLEDs) exhibit electroluminescence with fwhm < 40 nm, validating high-purity display compatibility. The simplified approach and mechanistic clarity advance the production of high-color-purity AIGS QLEDs, offering a paradigm for narrow emission in eco-friendly I–III–VI QDs.



Quantum dot light-emitting diodes (QLEDs) are being actively developed as a next-generation technology for highly energy-efficient solid-state lighting and displays. Quantum dots (QDs) serve as a critical component with remarkable properties, such as a wide color gamut, high color purity, excellent photoluminescence quantum yield (PLQY), and low-cost solution processing,^{1–4} which have contributed to the rapid development of energy-saving QLEDs.^{5–9} However, the lead/cadmium toxicity in conventional QDs limits their sustainable development. I–III–VI QDs, with direct bandgap,¹⁰ tunable spectrum¹¹ and high PLQY¹² are regarded as eco-friendly alternatives.^{13–18} Among them, Ag–In–Ga–S (AIGS) quaternary QDs have shown promise for lighting and electroluminescent devices due to their full-spectrum tunability^{19–22} and high PLQY (>80%).^{23,24} However, the defect-related broad emission spectra limit their use in high-definition displays.

The broad-emission I–III–VI QDs is attributed to intrinsic internal structural defects (such as V_{Cu} , V_{Ag} , V_{S} , Ag_i , In_{Ag} , Ga_{Ag} , etc.) participating in intraband charge carrier radiative recombination, including donor–acceptor pair (DAP) radiative recombination and band-to-bound radiative recombination.^{17,25–27} To narrow the emission, structural defects are often targeted. Kuwabata's group pioneered core–shell

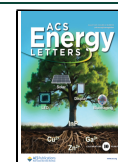
structures using amorphous GaS_x to passivate AgInS_2 QDs, achieving a narrow-emission QDs with a full width at half-maximum (fwhm) of 32 nm, demonstrating the feasibility of I–III–VI QDs in narrow emission for the first time.²⁸ Subsequent studies explored similar approaches for achieving narrow-emission AIGS QDs.^{29–36} Lee et al.³³ demonstrated the potential for narrow-emission AIGS QDs in the visible light spectra by constructing an AIGS/ AgGaS_2 core–shell structure. Heesun Yang et al.³⁵ designed a multishell structure AIGS/ $\text{AgGaS}_2/\text{Ga}_2\text{S}_3$ QDs to achieve narrow emission while suppressing the emission tail phenomenon. Despite progress, core–shell synthesis is complex, with poor reproducibility and high costs, limiting the large-scale production and application of AIGS QDs. Multistep injection synthesis and multishell structure also complicate the emission mechanism analysis.^{28,37,38} Achieving narrow emission in uniform alloyed AIGS QDs could effectively circumvent the aforementioned

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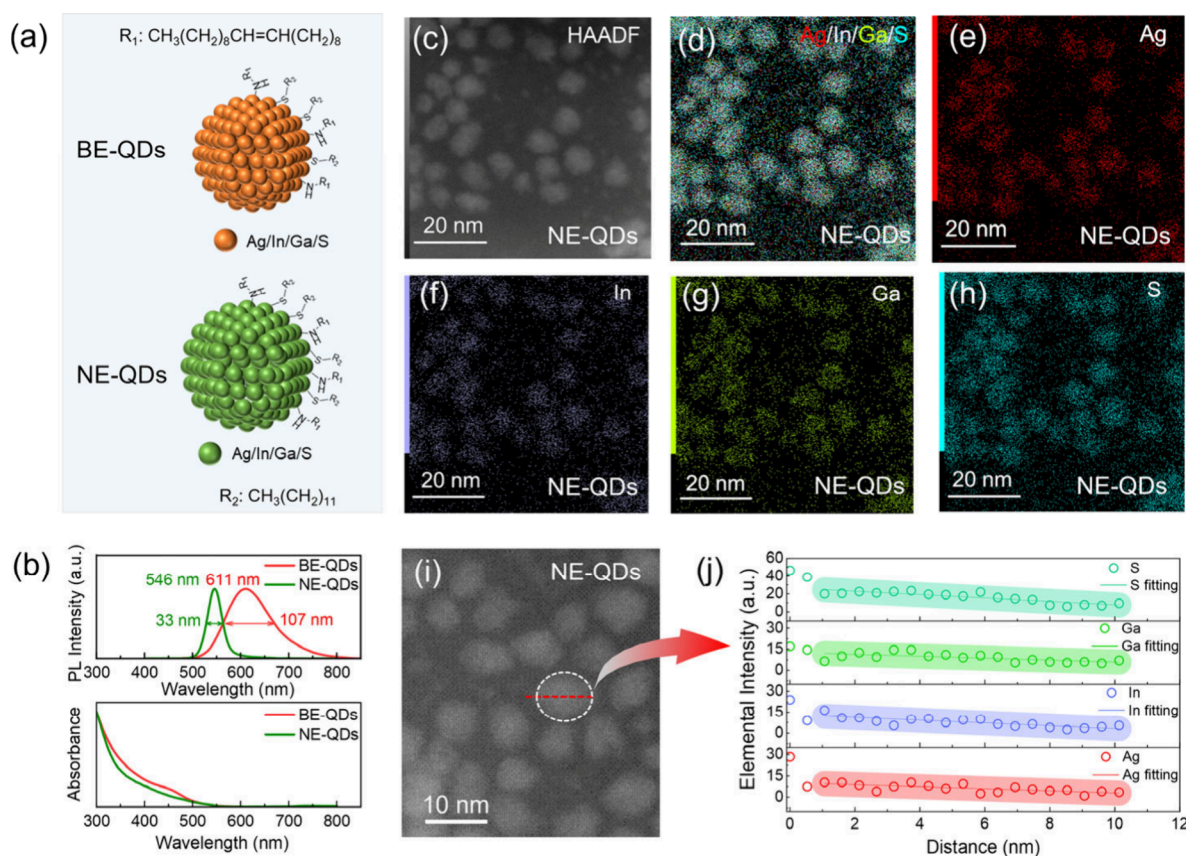


Figure 1. (a) Schematic illustration of BE-QDs and NE-QDs. (b) PL spectra (top) and UV-vis absorption (bottom) of AIGS QDs. (c) STEM-HAADF image and (d–h) EDX elemental mapping of NE-QDs. (i, j) EDX linear scanning results of NE-QDs.

challenges. Simplifying the synthesis method and disassembling the growth process for analysis would be an effective way to promote efficient synthesis and emission mechanism analysis of narrow-emission AIGS QDs.

Here, we developed a one-pot method for synthesizing narrow-emission AIGS QDs with a uniform alloyed structure via defect self-compensation. This approach simplifies synthesis and allows for a detailed growth process analysis. We obtained AIGS QDs with emissions from 525 to 595 nm, exhibiting fwhm's range of 30–38 nm and high PLQYs. A typical 595 nm-emission QD demonstrated a representative fwhm of 30 nm and a PLQY of 95.9%. QLEDs based on these narrow-emission QDs also demonstrated electroluminescence (EL) emission with a fwhm below 40 nm. Microstructural analyses revealed that these AIGS QDs possessed an alloyed structure by uniform elemental distribution, contrasting with a classic core–shell structure. The narrow emission was due to reduced structural defects through Ag_i self-compensation to V_{Ag} . During synthesis, the unreacted Ag compensated for some of the V_{Ag} , and meanwhile the Ag_i migrated to fill V_{Ag} under high temperature. This synergy between these processes facilitated the realization of the final narrow-emission AIGS QDs. Our simplified synthesis of narrow-emission AIGS QDs lays the foundation for the preparation of high-color-purity QLEDs. Analyzing the growth process also provides new insights into the luminescence mechanism of AIGS QDs, offering a reference for future research on I–III–VI QDs.

As illustrated in Figure S1a, we synthesized AIGS QDs based on the upgrade and improvement of the synthesis method previously reported by our group.²⁴ Fluorescence evolution

during growth is shown in Figures S1b,c and S2. Here, QDs obtained at low temperatures (150 °C) are defined as BE-QDs, and those obtained at high temperatures (280 °C) are defined as NE-QDs. The schematic diagrams of QDs are shown in Figure 1a. Photoluminescence fluorescence (PL) spectra (Figure 1b, top) indicated that BE-QDs exhibited a broad emission peak at 611 nm with a fwhm of 107 nm, whereas NE-QDs displayed a narrow emission peak at 545 nm with a fwhm of only 33 nm. Transmission electron microscopy (TEM) analysis (Figure S3) revealed an increase in size from 3.06 nm (BE-QDs) to 7.53 nm (NE-QDs), suggesting that the PL blue shift was not due to the quantum confinement effect. Absorption spectra (Figure 1b, bottom) showed that the absorption bands of BE-QDs and NE-QDs were essentially consistent, with bandgap energy (E_g) of 2.30 and 2.27 eV respectively (Figure S4a). The bandgap energies were comparable to the PL emission center energy of NE-QDs, implying that the blue shift and narrowing of PL spectra were induced by the transformation of the luminescence mechanism of QDs. X-ray diffraction (XRD) results (Figure S4b) revealed that both BE-QDs and NE-QDs had diffraction peaks matching $AgInS_2$ (PDF#75-0117) and $AgGaS_2$ (PDF#75-0114) alloys,³⁹ and the sharper peaks in NE-QDs suggested a higher crystallinity. It was speculated that this may be related to the reduction of structural defects of AIGS QDs, which will be further discussed and analyzed in the following research.

Scanning transmission electron microscopy-high angle annular dark field imaging (STEM-HAADF) and energy-dispersive X-ray spectroscopy (EDX) mapping (Figures 1c–h and S5) revealed uniformly distributed Ag, Ga, In, and S

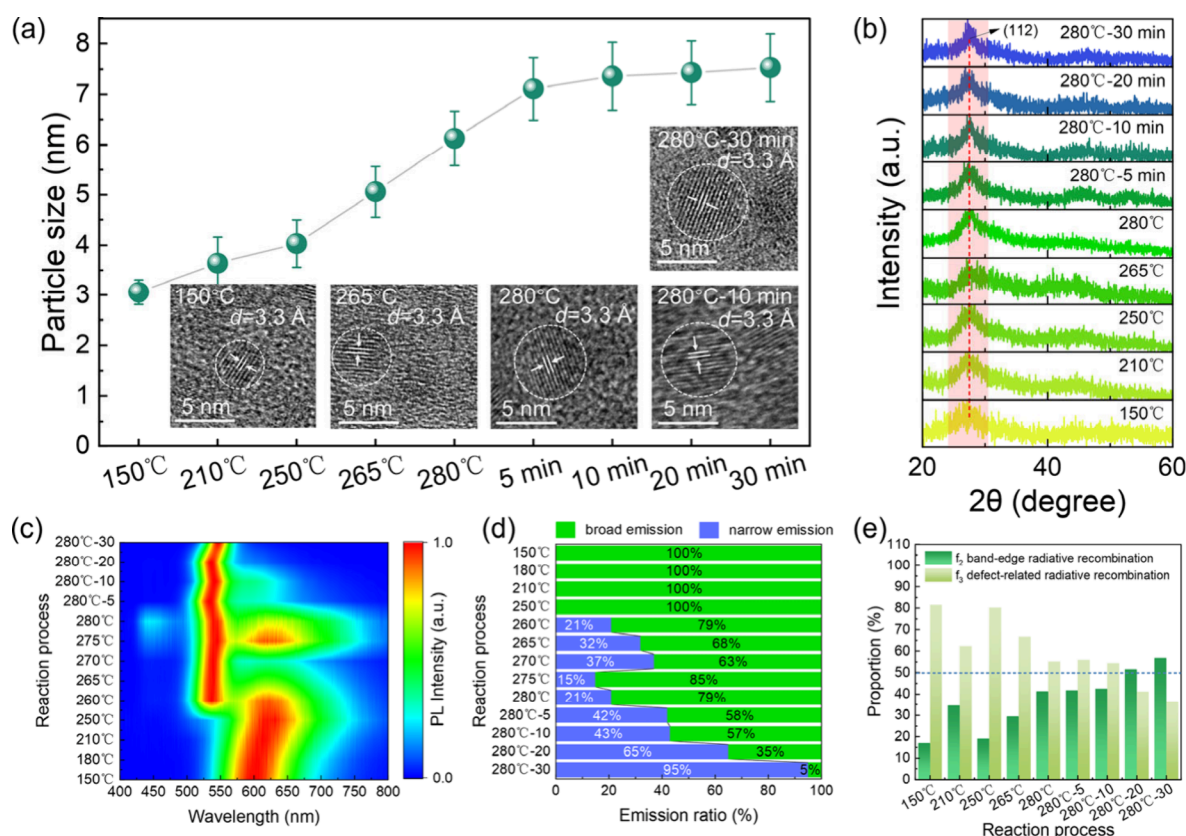


Figure 2. (a) Average particle size evolution, (b) XRD evolution, and (c) PL intensity color mapping of AIGS QDs during the growth. (d) Gaussian deconvolution results of PL spectra. (e) The proportion of different recombination mechanisms of AIGS QDs analyzed by TRPL spectra.

signals without local aggregation/absence in both QDs, preliminarily confirming their alloyed (noncore–shell) structure. This uniformity was further supported by XPS depth profiling (Figure S6 and Table S1), showing stable elemental ratios across etching depths. Elemental line-scanning of single NE-QDs (Figures 1i,j) demonstrated a consistent distribution internally and externally. Additionally, lattice fringes in STEM-HAADF images (Figure S7a) were continuous and uniform with FFT/iFFT analysis (Figure S7b–d) confirming the uniformity of the crystal plane spacing. These results collectively demonstrate that NE-QDs adopt a homogeneous alloyed structure rather than a core–shell configuration. Consequently, narrow emission in our work likely originates from modified internal defects in QDs, contrasting with prior reports attributing narrow emission to core–shell passivation.^{32,40}

Comprehensive growth analysis was imperative to elucidate the narrow emission mechanism of alloyed AIGS QDs. Figures 2 and S8–S10 illustrate the evolution from broad to narrow emission. During the temperature-ramping stage (150 to 280 °C), the QDs size increased from 3.06 to 6.13 nm, accompanied by PL redshift and widening (Figures 2a,c, S8 and S10). While the absorption tail also exhibited a consistent redshift phenomenon (Figure S9a), which is attributed to the quantum size effect. XRD (Figure 2b) revealed stable diffraction peaks despite temperature changes, suggesting a preserved crystal structure during the epitaxial regrowth process, which was attributed to temperature-driven redispersion and readsorption of unreacted precursors onto existing QD surfaces. Consistent interplanar spacing at different

temperatures (Figure 2a, inset) further confirmed the structural stability. During the holding stage at 280 °C, QDs sizes increased slowly and gradually stabilized over time, while XRD showed crystallinity improvement, potentially attributable to a reduction in internal structural defects within QDs. Urbach energy (E_u) calculations (formulas S1 and S2) substantiate this. E_u reflects the density of localized states near the bandgap; a higher E_u signifies greater defect density, while a lower E_u indicates reduced tail states arising from defects or lattice distortions.^{41,42} As shown in Figure S9b–g, the E_u diminished from 0.257 eV (280 °C-0 min) to 0.046 eV (280 °C-30 min) as time increased, confirming a significant reduction in localized states density and verified the decline in structural defect density within AIGS QDs when maintained at 280 °C for 30 min.

A more comprehensive understanding of the growth of NE-QDs can be obtained by analyzing the evolution of the PL emission. Figures 2c and S10a showed that during the 150–280 °C heating, the fwhm's of QDs remained above 100 nm. At 260 °C, narrow green emission emerged and accounts for 21% of total emission (Figure 2d), signaling a new transition from defect to band emission in AIGS QDs. Considering the defect variation analyses, it could be inferred that temperature-driven structural defect changes within QDs created a new radiative recombination channel, which became more dominant with increasing temperature. The anomalous reduction at 275 °C was attributed to the thermal quenching effect at high temperatures, where low-activation-energy defects trapped excitons, suppressing band-edge transitions.⁴³ By 280 °C, narrow emission rebounded, and an extended reaction time

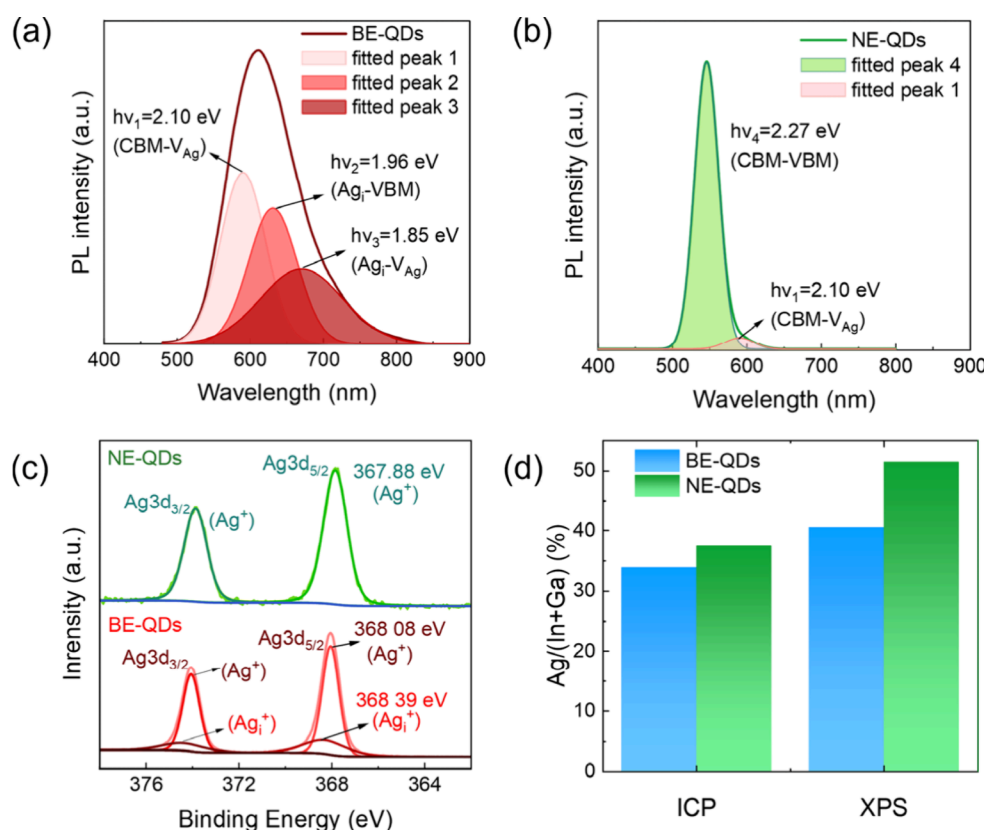


Figure 3. Gaussian deconvolution of the PL spectra of (a) BE-QDs and (b) NE-QDs. (c) High-resolution XPS spectra of Ag 3d and (d) the elemental ratio of Ag/(In+Ga) in BE-QDs and NE-QDs.

triggered competition between band-edge and defect-related emissions. As the holding time increased, narrowed fwhm as red emission (600–700 nm) diminished, while green emission (500–600 nm) intensified. After 30 min, QDs transformed from bimodal to unimodal emission, with fwhm decreasing to 33 nm and narrow emission proportion increasing from 21% to 95%, suggesting that band-edge emissions dominated QDs at this stage.

Complementary TRPL analysis (fitted via formulas S3 and S4) identified the carrier dynamics of QDs, and the results are summarized in Table S2. The faster one (τ_1) corresponds to nonradiative recombination, while the intermediate component (τ_2) represents band-edge radiative recombination. The slowest decay time (τ_3) is defect-related radiative recombination, including the DAP and band-to-bound radiative recombination in the I–III–VI QDs.^{23,44} As depicted in Figure S10b, in the broad emission dominated stage, AIGS QDs possessed a long carrier lifetime, reaching hundreds of nanoseconds. With an increase in the contribution of narrow emission, the carrier lifetime shortened to 64.10 ns. According to formula S5, the contribution ratios of each recombination mechanism was calculated (Figure 2e and Table S2). The comparison revealed that long carrier lifetimes correspond to high defect-related radiative recombination contributions. At 150–280 °C, defect-related radiative recombination contributions were over 50%, indicating broad emission originated from defect-related emission. As band-edge radiative recombination strengthened and dominated, the carrier lifetime significantly decreased and the corresponding PL emission peak was dominated by narrow emission, further again

demonstrating that narrow emission is due to band-edge radiative recombination.

According to the previous first-principles calculations report of AIGS QDs,⁴⁵ Ag-related defects with low formation energy (V_{Ag} , Ag_{In} , Ag_{Ga} , In_{Ag} , and Ga_{Ag}) play a pivotal role in the luminescence. Shallow-level defects V_{Ag} and Ag_i located near the valence band maximum (VBM) and conduction band minimum (CBM), respectively, can act as carrier providers and contribute to defect-related radiative recombination. Gaussian deconvolution of BE-QDs revealed that the broad-emission spectrum could be well fitted to three emission peaks at 2.10, 1.96, and 1.85 eV (peak 3), as shown in Figure 3a. In broad-emission AIGS QDs it is generally believed that V_{Ag} is involved in radiative recombination as an acceptor defect while Ag_i is a donor defect.^{45,46} According to defect energy level positions and deconvolution analysis, it could be considered that the three emission peaks in BE-QDs corresponded to three radiative recombination channels, corresponding to CBM- V_{Ag} (peak 1), Ag_i -VBM (peak 2), and Ag_i - V_{Ag} (peak 3), respectively. Interestingly, in NE-QDs, the intensity of peak 1 exhibited a sharp decline, while the intensities of peaks 2 and 3 became undetectable. Instead, a high-intensity narrow emission peak (defined as peak 4) appeared at 546 nm, as presented in Figure 3b. Based on the analysis presented in Figure 3c, the emission area of peak 4 accounted for 95%, suggesting that Ag_i and V_{Ag} were significantly reduced in NE-QDs compared with BE-QDs. It was reasonably speculated that it was a consequence of the effective compensation of silver vacancies in the AIGS QDs.

To further support the inference of changes in Ag-related defects, the Ag valence in both AIGS QDs was discussed by X-

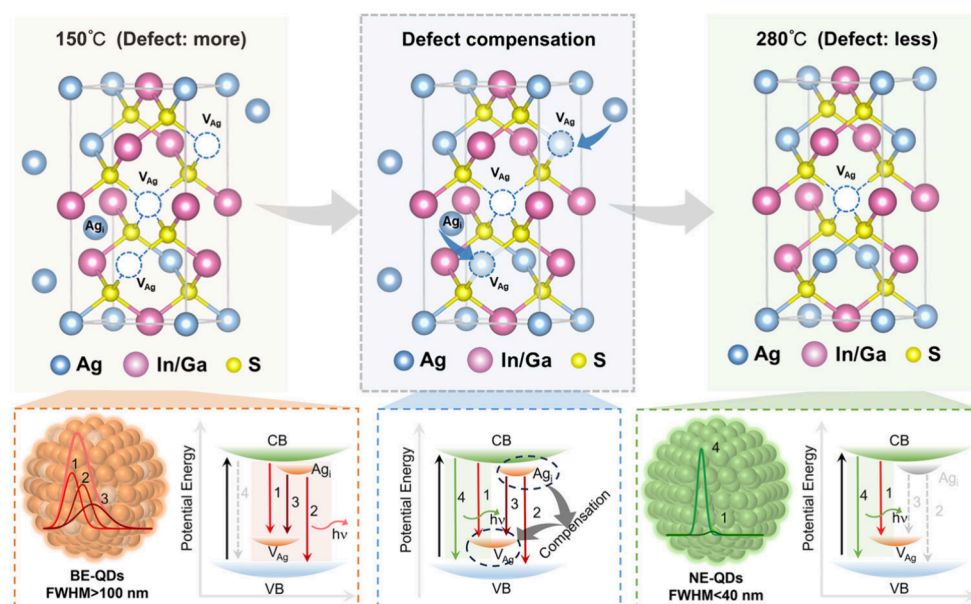


Figure 4. Illustration of the mechanistic model of the narrow emission implementation of AIGS QDs.

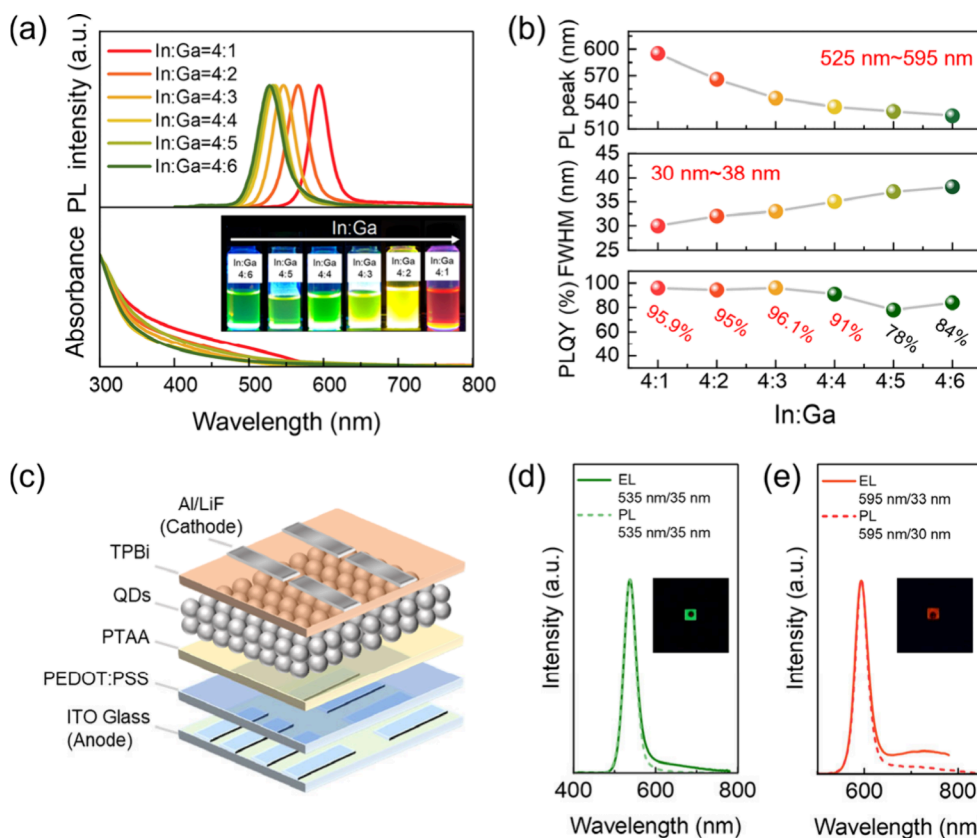


Figure 5. (a) PL and UV-vis absorption spectra (inset, photographs of the corresponding NE-QDs under UV light), (b) detailed optical performance, (c) QLED device structure, and (d, e) EL and PL spectra of QLEDs based on NE-QDs.

ray photoelectron spectroscopy (XPS). The XPS spectra of QDs are collected in Figure S11a. When Ag^+ exists as a donor defect (Ag_i^+) in the unexcited QDs, it is occupied by electrons and is electrically neutral so that it can be regarded as electrically neutral Ag^0 . As reported in previous studies, Ag^0 has a higher binding energy compared to Ag^+ under normal conditions.^{47,48} For BE-QDs, the high-binding-energy orbital peaks (368.6 and 374.6 eV) associated with Ag_i^+ were observed

as demonstrated in Figure 3c.⁴⁹ These peaks disappeared in NE-QDs, leaving only the low-energy orbital peak for monovalent Ag (Ag^+), indicating a significant reduction of Ag_i defects in NE-QDs compared to BE-QDs. According to the ICP data and XPS elemental content analysis (Tables S3 and S4 and Figure 3d), NE-QDs had a higher proportion of Ag than did BE-QDs. Consequently, combined with the previous Gaussian deconvolution analysis of the PL spectra, it could be

concluded that, in NE-QDs, on the one hand, the elevated Ag content compensated for the Ag vacancies in the structure. On the other hand, the originally existing Ag_i atoms in the QDs migrated under high-temperature driving to fill the Ag vacancies. The synergistic action of both resulted in a reduction in V_{Ag} and Ag_i defects in QDs. Furthermore, XPS and electron paramagnetic resonance analysis have ruled out that sulfur vacancies (V_s) might play a role in narrow emission, the results were shown as Figure S11b,c.

Ultraviolet photoelectron spectroscopy (UPS) analysis provided supplementary evidence of the changes in Ag-related defects, as shown in Figure S12. The Fermi level (E_F) of BE-QDs was closer to the VBM, demonstrating p-type characteristics, as V_{Ag} can serve as a hole carrier and provide a large number of hole carriers.⁴⁵ In NE-QDs, the p-type characteristics of the QDs were weakened, and the E_F shifted away from the VBM. This suggested that compared to BE-QDs, the hole carriers' concentration in NE-QDs diminished, aligning with the result of V_{Ag} reduction analyzed above.

Based on the discussion above, we concluded that the narrow emission of uniform alloyed AIGS QDs is achieved by defect compensation, as depicted in Figure 4. In BE-QDs (150 °C), numerous V_{Ag} and Ag_i defects provided multiple carrier radiative recombination channels (solid line part: channels 1, 2, 3), suppressing the intrinsic band-edge emission process and causing the broad-spectrum emission (fwhm >100 nm). When the reaction temperature increased from 150 to 280 °C, V_{Ag} defects would be effectively compensated by the unreacted Ag_i^+ and the Ag_i existing in the lattice. The channel 1 (CBM- V_{Ag}) was significantly inhibited, channel 2 (Ag_i -VBM) and channel 3 (Ag_i - V_{Ag}) were closed due to the disappearance of Ag_i defect. Consequently, the intrinsic band-edge radiative recombination process (channel 4) regained its competitiveness, and the QDs exhibited narrow emission with a fwhm of less than 40 nm. Overall, the narrow spectral emission of the AIGS QDs is achieved through defect compensation. The radiative recombination path returned to the band-edge recombination, realizing the correction of the luminescence color purity of the QDs.

Furthermore, by adjusting the In:Ga, the spectral coverage of the alloyed NE-QDs was expanded without enhancing defect-related emissions, maintaining narrow emissions. As Ga content increased, the absorption and emission spectra of these QDs underwent a blue shift, due to an enlargement of the E_g (Figure S13a). The PL spectra gradually shifted from 595 to 525 nm as Ga increased, presenting visible fluorescence ranging from orange-red to green (Figure 5a). Notably, the PL spectra of the NE-QDs with different components had narrow fwhm's range of 30–38 nm, with a minimum value of 30 nm at In:Ga = 4:1 as depicted in Figure 5b, which is a recorded value compared with previously reported quaternary alloyed AIGS QDs (Table S5), and all these NE-QDs demonstrated considerable PLQY (78%–96.1%). Additionally, the photon energy of the PL emission center (E_{PL}) of NE-QDs with different components was almost equivalent to that of the E_g , suggesting that all QDs still contributed to luminescence via band edge radiative transitions (Figure S13b). TRPL analysis (Figure S14 and Table S6) reconfirmed the dominant contribution of the band-edge radiative recombination process (f_2), and the XRD with alloyed structure characteristics (Figure S15) also reinforced that the narrow emissions were not contributed by the core-shell structure.

Taking In:Ga = 4:4 and 4:1 as representative examples, corresponding AIGS QLED devices (structure shown in Figure

5c) were fabricated to evaluate the electroluminescent performance. The narrow electroluminescence fwhm (≤ 40 nm) enables a wider color gamut and improved color accuracy, aligning with the stringent requirements of next-generation high-purity displays such as the Rec. 2020 standards. As depicted in Figure Sd,e, these QLEDs exhibited EL spectra consistent with PL, displaying high-color-purity electroluminescence of green and orange-red with fwhm's of 35 and 30 nm, respectively. Electrical performance characterization (Figure S16) revealed that these QLEDs exhibited low turn-on voltages of 2.8 and 3.2 V, and showed a peak luminance of 47.8 and 19 cd/m^2 and maximum external quantum efficiency (EQE) of 0.60% and 0.12%, respectively. The EQE performance is at an advantageous level compared with the previously reported narrow-emission AIGS QLEDs (Table S7). Although there is room for performance enhancement compared to established QD systems, this study highlights the potential of uniform alloyed AIGS QDs in high-color-purity electroluminescent applications. Our synthesis protocol achieves a balance between exceptional color purity and competitive PLQY, positioning AIGS QDs as viable candidates for next-generation display technologies. Future work will focus on optimizing device architecture and performance parameters through device engineering.

In conclusion, we developed a method for synthesizing narrow-emission AIGS QDs with a uniform alloy structure via defect self-compensation, covering emission from 525 to 595 nm. These QDs exhibited narrow fwhm's range of 30–38 nm and high PLQY. Notably, the 595 nm-emission QDs had an impressive fwhm of 30 nm and PLQY of 95.9%. The resulting QLED devices showed high color purity, with EL emission fwhm matching PL. Research results revealed that the narrow emission stemmed from the minimized Ag-related structural defects due to the defect self-compensation of Ag_i to V_{Ag} . The defect reduction closed wide-spectrum emission channels and enhanced band-edge emission, achieving high-purity emission. This work provides a strategy for alloyed, narrow-emission AIGS QDs, offering new references for narrow-emission I–III–VI QDs and advancing the sustainable development of high-purity I–III–VI QLED, reducing the reliance on toxic elements in future electronic products.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsenerylett.5c01454>.

Materials, synthesis methods and QLED fabrication of AIGS QDs; Photographs of AIGS QDs solution, TEM, HRTEM, and EDX of BE-QDs and NE-QDs; Bandgaps, XRD, relationship of $\ln(A)$ vs $h\nu$, XPS, EPR and TRPL of AIGS QDs; Table of fitted results of TRPL and table of atomic ratios of AIGS QDs (PDF)

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Notes

The authors declare no competing financial interest.

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